Reg No.:_______________ Name:__________________________

APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY

Sixth Semester B.Tech Degree Examination June 2022 (2019 Scheme)

Course Code: EET322

Course Name: RENEWABLE ENERGY SYSTEMS

Max. Marks: 100 Duration: 3 Hours

PART A

Answer all questions, each carries 3 marks.

			Marks
1		Discuss the importance of Kyoto protocol in sustainable development.	(3)
2		Explain the consequences of Global warming.	(3)
3		Differentiate between concentrating and non-concentrating type solar collectors.	(3)
4		Illustrate how solar PV array is realised using PV panels and PV modules.	(3)
5		List the factors that influence the site selection for wind power plant.	(3)
6		Briefly explain the concept of power coefficient and Betz limit with reference to a wind turbine.	(3)
7		Explain the term “tidal range” with the help of a figure.	(3)
8		Explain biofouling with reference to OTEC power plants.	(3)
9		Explain how hydrogen is stored using metal hydride.	(3)
10		Write short notes on Refuse Derived Fuel.	(3)

PART B

Answer one full question from each module, each carries 14 marks.

Module I

11	a)	Discuss the relationship between energy and sustainable development.	(5)
	b)	List the various conventional sources of energy and explain their advantages and disadvantages.	(9)

OR

12	a)	Explain the need to increase the utilization of non-conventional energy sources.	(6)
	b)	Explain briefly the causes and effects of pollution.	(4)
	c)	List the various pollutants and explain their harmful effects.	(4)

Module II

13	a)	With neat diagram, explain the construction and of a liquid flat plate solar collector.	(6)
	b)	Explain maximum power point tracking in a solar PV system using buck-boost converter.	(8)

OR

14	a)	Explain the construction and working principle of a Pyranometer with neat diagram.	(8)
	b)	With neat block diagram, explain the grid-interactive or grid tied photovoltaic system.	(6)

Module III

15	a)	With neat diagrams, differentiate between Horizontal Axis Wind Turbine and Vertical Axis Wind Turbine. Also, compare their relative advantages and disadvantages.	(8)
	b)	Explain the general block diagram of a grid connected Wind Energy Conversion system.	(6)

OR

16	a)	Using relevant diagram, explain the terms pitch angle, angle of attack, solidity, relative velocity, lift force and drag force in the context of wind energy conversion systems.	(8)
	b)	Explain the output power versus wind speed characteristics of a wind turbine.	(6)

Module IV

17	a)	List three advantages and disadvantages of tidal power plants.	(6)
	b)	With a neat diagram explain hybrid cycle OTEC power plant. Also, list any two advantages of OTEC system.	(8)

OR

18	a)	With a neat diagram explain closed cycle OTEC power plant.	(6)
	b)	Classify tidal power plants based on the type of basin used and explain their working with relevant figures.	(8)

Module V

19	a)	List the factors that affect biogas generation.	(6)
	b)	Explain the working of Proton-Exchange Membrane Fuel Cells (PEMFC) with neat diagram.	(8)

OR

20	a)	With neat diagram, explain how power can be extracted using satellite stations.	(6)
	b)	With a neat diagram, explain the working of KVIC model/floating drum biogas plant.	(8)

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